A Multichannel Hybrid 2D–3D-CNN for Hyperspectral Image Classification With Small Training Sample Sizes

Shih-Yu Chen, *Member, IEEE*, Po-Yu Chu, Kuan-Liang Liu, and Yu-Cheng Wu

***Abstract*— This article presents an innovative multichannel hybrid 2D–3D-convolutional neural network (MH-2D-3D-CNN) model specifically designed for the challenging task of hyperspectral image classification (HSIC) with small labeled training sample sizes (SLTSSs). By integrating three-channel network architectures and incorporating target detection and attention mechanisms with 3D- and 2D-CNN techniques, our model emphasizes the extraction and fusion of spectral and spatial features, essential for accurate classification. In the first channel, we employ the normalized adaptive matched detector (NAMD) to enhance spectral features, enabling the extraction of intricate spectral information. Concurrently, 2D-CNN is applied to capture spatial features, enhancing the understanding of spatial structural information within the images. In the second channel, the convolutional block attention module (CBAM) is employed alongside 2D-CNN to further enhance the saliency of spatial features. The third channel adopts a concatenated approach, combining 3D-CNN and 2D-CNN to extract spectral and spatial features simultaneously, reinforcing spatial feature extraction capabilities. The merged feature maps from the three channels undergo final classification through fully connected layers, enhancing the model's stability in feature extraction and learning performance. Experimental validation was conducted on three widely used public datasets (Indian Pines [IP], University of Pavia [PU], and Salinas [SA]) with a fixed 1% labeled training sample size. The results demonstrate remarkable overall accuracy (OA) scores of 87.49%, 98.13%, and 99.22%. These results outperform state-of-the-art methods and showcase significant advancements, particularly in SLTSS scenarios.**

***Index Terms*— 2-D–3-D convolutional neural network (CNN), deep learning, feature extraction, hyperspectral image classification (HSIC), spectral–spatial analysis.**

I. Introduction

HYPERSPECTRAL imagery (HSI), characterized by intricate spectral signatures spanning numerous finely tuned wavelengths, possesses formidable material discrimination capabilities [1], [2]. As a result, hyperspectral imaging technology has found widespread utility across domains, including land use analysis, resource exploration [3], [4], environmental monitoring [5], [6], industrial production [7], and precision agriculture [8], [9], [10], [11].

The fundamental aim of hyperspectral image classification (HSIC) is pixel-wise precision categorization. To achieve this, a variety of statistical learning methods have been introduced for hyperspectral image analysis. While techniques, such as support vector machines (SVMs) [12], random forests (RFs) [13], and K-nearest neighbors (KNNs) [14], have the potential for hyperspectral image processing. However, the intricate high-dimensional and nonlinear nature of hyperspectral data often hampers the attainment of optimal classification outcomes. To elevate performance, data dimensionality reduction and feature extraction are commonly employed.

Recent advancements in deep learning have significantly enhanced the automatic extraction of deep and abstract features, leading to improved classification accuracy. Convolutional neural networks (CNNs), with their multilayered hidden structures, have emerged as successful tools in HSIC, enabling more efficient capture of both spectral and spatial features [15]. For instance, Makantasis et al. [16] and Chen et al. [17] have employed 2D-CNN and 3D-CNN for feature extraction from hyperspectral images and validated their classification performance. Hybrid models, such as 3-D–2-D-CNN [18] and HybridSN [19], which integrate both 3D-CNN and 2D-CNN architectures, have demonstrated enhanced efficacy. However, while these models deepen feature extraction, they also increase the complexity of training and require larger datasets, which can be a limitation in scenarios with small labeled training sample sizes (SLTSSs).

Since the introduction of CNNs in HSI, addressing SLTSS has become a critical challenge in HSIC. When sample availability is limited, fully leveraging CNNs' capabilities to extract both spectral and spatial features becomes essential for improving classification performance. This article tackles this challenge by proposing a novel method designed specifically to overcome sample scarcity while meeting practical application requirements. The key contribution of this article is the integration of advanced techniques into a multichannel architecture optimized for efficient feature extraction under SLTSS conditions. The normalized adaptive matched detector (NAMD) module enhances spectral feature extraction, the convolutional block attention module (CBAM) refines spatial saliency, and the fusion of 2D- and 3D-CNN ensures comprehensive feature learning. This combination yields significant performance gains across multiple datasets.

Received 22 June 2024; revised 7 October 2024; accepted 3 December 2024. Date of publication 9 December 2024; date of current version 13 December 2024. This work was supported in part by the "Intelligent Recognition Industry Service Center" through the Featured Areas Research Center Program under Higher Education Sprout Project by the Ministry of Education (MOE), Taiwan; in part by the National Science and Technology Council, Taiwan, under Grant 112-2622-E-224-003 and Grant 112-2221-E-224-037; and in part by Isuzu Optics Corporation. *(Corresponding author: Shih-Yu Chen.)*

The authors are with the Department of Computer Science and Information Engineering, National Yunlin University of Science and Technology, Douliu, Yunlin 64002, Taiwan (e-mail: sychen@yuntech.edu.tw).

Digital Object Identifier 10.1109/TGRS.2024.3512869

1558-0644 © 2024 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.

In Section III, we will further highlight the novelty of our approach, specifically focusing on how the integration of these techniques constitutes a significant advancement in HSIC with limited labeled data.

This article is organized as follows. Section II reviews HSIC methods for small training sample sizes, highlighting key challenges. Section III introduces the multichannel hybrid 2D–3D-CNN (MH-2-D-3-D-CNN) model, explaining its architecture and techniques for extracting both spectral and spatial features. Section IV outlines the experimental setup, datasets, and evaluation metrics. Section V presents the findings and compares them to state-of-the-art methods. Section VI examines model convergence, complexity, and parameter effects. Finally, Section VII summarizes the key insights and offers recommendations for future research.

II. Related Works

Challenges emerge when dealing with SLTSS problem, including the risk of overfitting. As highlighted by Gu et al. [20], models often excel on training data but underperform on novel data. The complexity of learning intricate features increases due to limited samples, which impedes the ability to encompass diverse variations. Training deep networks is also challenging because deep learning models typically require substantial sample sizes. In addition, feature selection becomes more complex when samples are scarce, and potential class imbalances can skew models toward predicting more abundant classes.

In hyperspectral image processing, these challenges are intensified by the high dimensionality of hyperspectral data, often referred to as the "curse of dimensionality," which increases the likelihood of overfitting with limited samples and constrains the effectiveness of conventional classification and analysis methods. The limited number of labeled samples further exacerbates the problem, making it difficult to train deep learning models effectively and increasing the risk of overfitting. To address these issues, numerous approaches targeting SLTSS have been proposed.

Li and Qian [21] utilized sparse nonnegative matrix factorization (sparse NMF) for selecting spectral bands, which helps manage complex spectral information. Cai et al. [22] employed a band attention module (BAM) and a reconstruction network (RecNet) for band selection, assisting in handling complex spectral data. Fu et al. [23] integrated band selection and spatial denoising to extract spectral features while using enhanced 2-D singular spectrum analysis (E2DSSA) to reduce spatial noise. Wang et al. [24] developed a multiscale feature fusion network based on global weighting (MSFGW) to capture multiscale spatial–spectral information, and Dong et al. [25] employed a BAM combined with CNN for classification. Zhao et al. [26] introduced a compact band weighting (CBW) lightweight module to reduce model complexity, while Hu et al. [27] developed a lightweight 2-D convolutional long short-term memory unit, named TT-ConvLSTM2D, using tensor-train decomposition (TTD) to compress training parameters. This was applied to two state-of-the-art ConvLSTM2D-based HSIC models.

Further advancements include Hu et al. [28] who proposed a lightweight tensor attention-driven ConvLSTM neural network (TACLNN), constructing a learnable tensor attention residual block (TARB) to enhance geometric structures and recover accuracy loss. Wang et al. [29] introduced the GloRe-HCNN model that extracts spatial and spectral features of hyperspectral data through 3D-CNN and 2D-CNN, utilizing the spatial–spectral global reasoning unit (SS-GloRe-Unit) for in-depth feature insights. Wang et al. [30] emphasized the importance of robust feature extraction for subsequent integration, highlighting feature selection, extraction, and amalgamation to improve classification models. Li et al. [31] introduced the MFDN model, utilizing principal component analysis (PCA) for dimensionality reduction to extract low-dimensional hyperspectral image data, combining 2D-CNN and 3D-CNN to mitigate the vanishing gradient problem and enhance pivotal features. Xiang et al. [32] proposed a concatenated 3D-CNN and 2D-CNN model for HSIC, incorporating a dense 3D-CNN for spectral–spatial feature extraction with a coordinate attention mechanism for classification.

Recent work by Dosovitskiy et al. [33] marked a shift in computer vision with the vision transformer (ViT), which uses a self-attention mechanism to capture global features of images and has been effectively adapted for hyperspectral image processing, where managing complex spectral–spatial relationships is critical. Building on this, Hong et al. [34] introduced SpectralFormer, integrating transformers with an efficient spectral–spatial fusion module to improve classification performance while preserving essential spectral information. Zheng et al. [35] introduced FreeNet, designed for fully end-to-end HSIC, bypassing traditional patch-based methods to enhance efficiency and performance. Ji et al. [36] advanced this field with PASSNet, a network incorporating a patch attention module to improve the extraction of spatial–spectral features and thus enhance the classification accuracy. Wang et al. [37] introduced a unified multiscale learning (UML) framework for HSIC, focusing on capturing important features across multiple scales to improve the performance. Mei et al. [38] proposed the group-aware hierarchical transformer (GAHT), applying a hierarchical attention mechanism that considers both spectral and spatial information for precise HSIC.

III. Proposed Approach

In the domain of hyperspectral data analysis, optimizing feature extraction and classification methodologies is pivotal.

This article presents an advanced MH-2D-3D-CNN architecture for HSIC with SLTSS. The model integrates multiple channels to leverage diverse processing techniques, aiming for improved efficiency and accuracy.

The proposed model comprises three distinctive channels, as depicted in Fig. 1.

1) Channel-1 integrates NAMD with a 2D-CNN architecture. Initially, the original images undergo processing through NAMD, which enhances the extracted target spectral features and reduces dimensionality. These refined features are subsequently input into a 2D-CNN, consisting of three convolutional layers followed by batch normalization, culminating in an additional convolutional layer and global average pooling [42]. This combination produces optimized feature vectors. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.

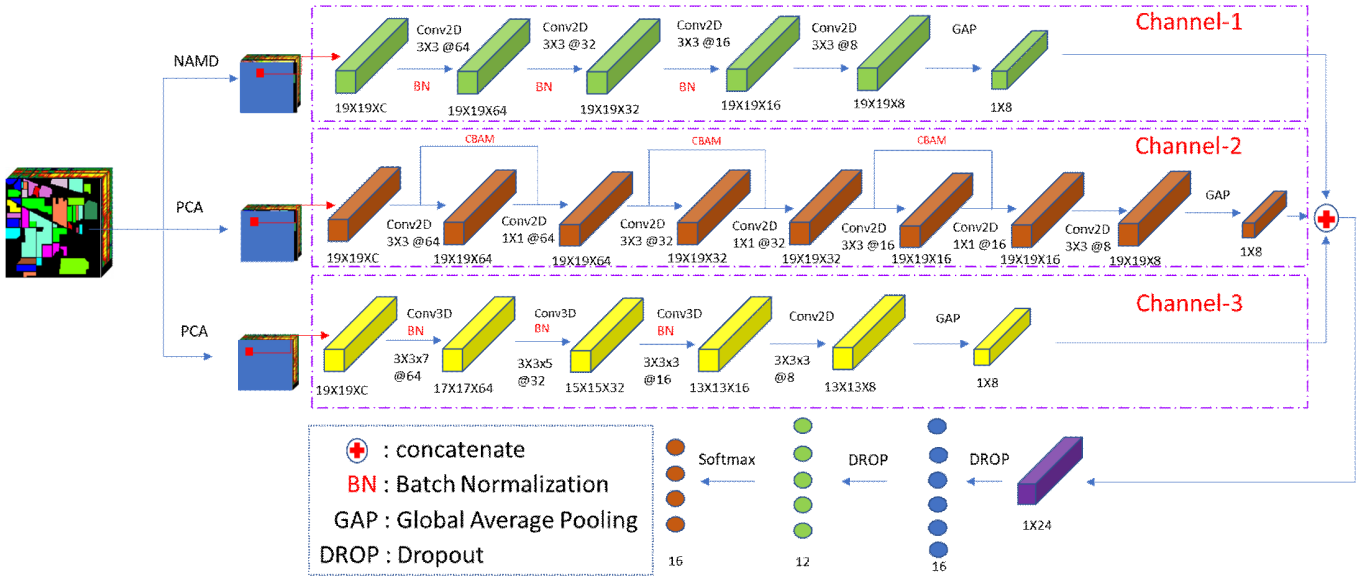


Fig. 1. Proposed MH-2D-3D-CNN architecture.

fusion of NAMD with the 2D-CNN proves particularly effective in HSIC, especially when confronted with limited sample sizes. It emphasizes target-specific features that traditional dimensionality reduction techniques, such as PCA, may overlook, thereby facilitating the identification of subtle spectral differences and enhancing overall classification accuracy.

2) Channel-2 and Channel-3 both apply PCA for dimensionality reduction and feature extraction. In Channel-2, the feature maps are refined using the CBAM. The outputs from this process are concatenated and subsequently processed through additional convolutional layers, followed by global average pooling. CBAM dynamically adjusts spatial and channel weights, enabling the model to concentrate on the most relevant features, which enhances the classification accuracy. This reliance on 2-D convolutions ensures computational efficiency, making Channel-2 well suited for balancing performance with processing speed.
3) Channel-3 employs a 3-D–2-D concatenated CNN architecture. The incorporation of 3-D convolutions is particularly effective in capturing both spatial and spectral features simultaneously, allowing this channel to extract rich, multidimensional information from hyperspectral images. Furthermore, the 2D-CNN component reinforces spatial feature extraction, enhancing the overall feature representation. This integrated approach proves particularly beneficial for handling high-dimensional data, enhancing classification accuracy in complex datasets. It is especially effective in scenarios where the interaction between spatial and spectral dimensions plays a critical role in achieving optimal performance.

In the final step, the feature vectors generated by the three channels are concatenated, forming a comprehensive and enhanced feature vector. Noise reduction is achieved through fully connected layers, and the final classification is performed using a softmax operation.

In summary, the refined MH-2D-3D-CNN architecture proposed in this article represents a significant advancement in feature extraction and HSIC under conditions of SLTSS. By synergizing distinct processing methods across multiple channels, this innovative framework demonstrates considerable promise as a valuable tool for hyperspectral data analysis.

A. Principal Component Analysis

PCA [43] is a widely used dimensionality reduction technique, particularly effective in managing high-dimensional data, such as HSI. PCA operates by linearly transforming the original data into a set of orthogonal dimensions known as principal components. These components are ranked based on the amount of variance that they capture from the original data.

In the context of HSI, each pixel is represented by a spectrum comprising multiple spectral bands, resulting in a highly dimensional dataset. PCA addresses this complexity by identifying the principal components that account for most of the variance within the spectral data. By selecting only the most significant components, PCA effectively reduces the dimensionality of the dataset while preserving its critical information.

This dimensionality reduction is particularly advantageous for tasks such as HSIC and target detection. One of the key strengths of PCA is its ability to filter out redundant and noisy information, which in turn enhances the performance of subsequent analysis and modeling processes.

B. Window Sliding Segmentation

According to [44], the size of the hyperspectral windowing significantly impacts image processing and classification results. It determines the range of features extracted, influencing the model's ability to capture both local details and global structures. Smaller windows provide more detailed local information, allowing for the capture of fine features but may sacrifice important contextual information. In contrast, larger windows preserve the overall structure but may blur fine details and average out local variations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.

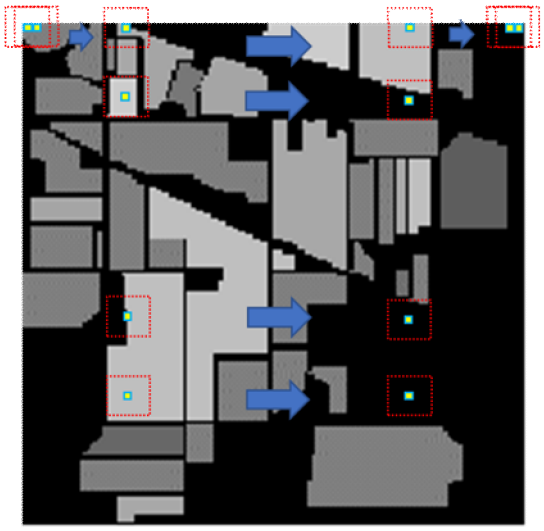


Fig. 2. Illustration of the window sliding segmentation.

Achieving a balance between local details and global structure is crucial when selecting the window size. The size should be tailored to the specific task and image characteristics to optimize feature extraction and classification performance. During the windowing process, pixel data surrounding a central pixel are extracted (as illustrated in Fig. 2), which helps to maintain local features and spatial relationships. In addition, zero padding can be applied to retain the feature map size and enhance classification accuracy.

C. NAMD Combined With 2D-CNN

NAMD [39] is an adaptive filter widely used for target detection and tracking.

The algorithm utilizes the statistical properties of Gaussian distribution to compute the similarity between the target signal and data samples, assigning a weight to each sample based on these similarities. NAMD generates the final output by performing a weighted average of these weights. Specifically, the algorithm compares the target signal with all data samples, assigning a weight to each sample according to their similarity. By using this weighted averaging method, NAMD can effectively extract key features of the target signal from the data samples, making it suitable for target detection and tracking tasks. NAMD is defined as

$$\delta^{\mathrm{NAMD}}(\mathbf{r}) = \frac{(\mathbf{t}-\boldsymbol{\mu}_\mathbf{b})^{\mathrm{T}}\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_\mathbf{b})}{(\mathbf{t}-\boldsymbol{\mu}_\mathbf{b})^{\mathrm{T}}\mathbf{K}^{-1}(\mathbf{t}-\boldsymbol{\mu}_\mathbf{b})} \tag{1}$$

where $\mathbf{t}$ represents the target signal; $\boldsymbol{\mu}_\mathbf{b}$ represents the mean of the background; $\mathbf{K}$ represents the covariance matrix of the background, capturing relationships between data samples; $\mathbf{r}$ represents the current data sample; $\mathbf{K}^{-1}$ represents the inverse of the covariance matrix; and $(\mathbf{t}-\boldsymbol{\mu}_\mathbf{b})^{\mathrm{T}}$ is the transposed difference between the target signal and the background mean, forming a row vector.

In this article, NAMD is applied to generate new feature images by selecting random points within each target class. This approach reduces the data dimension and enhances key spectral features, addressing sample scarcity and improving data diversity and model adaptability. After processing with NAMD, the images are input into a 2D-CNN, allowing the model to learn spatial features such as edges, textures, and shapes, which are crucial for classification.

By integrating NAMD with 2D-CNN, the model leverages both spectral and spatial information to improve HSIC. Spectral information reveals the optical properties of objects, while spatial information captures relationships within the image. This combination allows the model to develop a more comprehensive understanding of the image, leading to enhanced classification accuracy and overall performance. By capturing features at multiple levels, the model performs more effective feature extraction and classification, resulting in superior outcomes.

D. CBAM Combined With 2D-CNN

CBAM [40], [41] is an attention mechanism model that enhances the model's focus on key features by integrating the channel attention module (CAM) [45] and spatial attention module (SAM) [46]. CAM performs pooling and multilayer neural network combinations on the channel dimension of features to filter and weight features, making the model pay more attention to channel features that are crucial for classification while reducing the influence of redundant features. The formula for CAM is as follows:

$$\begin{aligned}\boldsymbol{M}_{\mathrm{c}}(\mathbf{F}) &= \sigma(\mathrm{MLP}(\mathrm{AvgPool}(\mathbf{F})) + \mathrm{MLP}(\mathrm{MaxPool}(\mathbf{F}))) \\ &= \sigma\left(\mathbf{W_1}\left(\mathbf{W_0}\left(\mathbf{F}^{\mathbf{c}}_{\mathbf{avg}}\right)\right) + \mathbf{W_1}\left(\mathbf{W_0}\left(\mathbf{F}^{\mathbf{c}}_{\mathbf{max}}\right)\right)\right)\end{aligned} \tag{2}$$

where $\boldsymbol{M}_{\mathrm{c}}(\mathbf{F})$ represents the channel attention map, AvgPool $(\mathbf{F})$ represents average pooling, MaxPool $(\mathbf{F})$ represents maximum pooling, MLP refers to a multilayer perceptron, and σ represents the sigmoid activation function. $\mathbf{W_0}$ and $\mathbf{W_1}$ are weight matrices applied in the MLP layers, and $\mathbf{F}^{\mathbf{c}}_{\mathbf{avg}}$ and $\mathbf{F}^{\mathbf{c}}_{\mathbf{max}}$ are the feature maps obtained after average and max pooling, respectively.

SAM, on the other hand, focuses on extracting spatial structural information from images by performing pooling and convolution operations on the spatial dimension, generating a spatial attention map to emphasize key spatial locations in the image. The formula for SAM is as follows:

$$\begin{aligned}\boldsymbol{M}_{\boldsymbol{s}}(\mathbf{F}) &= \sigma\left(f^{7\times7}\left(\left[\mathrm{AvgPool}(\mathbf{F}); \mathrm{MaxPool}(\mathbf{F})\right]\right)\right) \\ &= \sigma\left(f^{7\times7}\left(\left[\mathbf{F}^{\mathbf{s}}_{\mathbf{avg}}; \mathbf{F}^{\mathbf{s}}_{\mathbf{max}}\right]\right)\right)\end{aligned} \tag{3}$$

where $\boldsymbol{M}_{\boldsymbol{s}}(\mathbf{F})$ represents the spatial attention map; $f^{(7\times7)}$ is the 7×7 convolution kernel; AvgPool $(\mathbf{F})$ represents average pooling; MaxPool $(\mathbf{F})$ represents maximum pooling; σ represents the sigmoid activation function; and $\mathbf{F}^{\mathbf{s}}_{\mathbf{avg}}$ and $\mathbf{F}^{\mathbf{s}}_{\mathbf{max}}$ are the spatial feature maps obtained after average and max pooling, respectively.

In this section, we combine CBAM with 2D-CNN and apply it to a residual network [47]. In scenarios with limited sample sizes, 2D-CNN may face challenges such as insufficient training data and underutilization of spectral information. By introducing the attention mechanism of CBAM, the model automatically learns the relationships between features and focuses on key information. Meanwhile, a residual network can address the vanishing gradient problem, enabling the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.

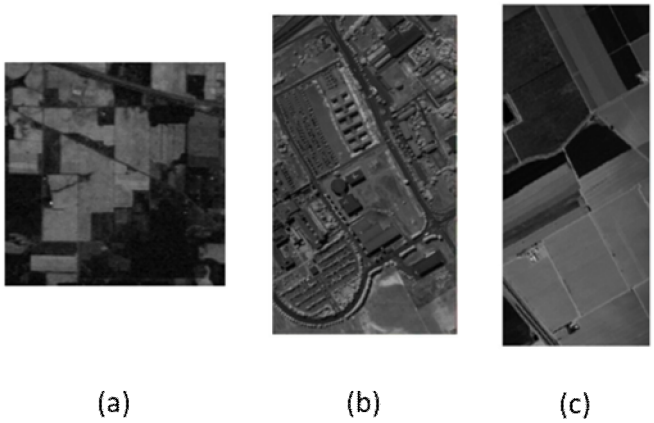


Fig. 3. Grayscale image of (a) IP, (b) PU, and (c) SA.

model to train deeper and enhance feature representation capabilities. This combination of CBAM and residual network improves feature extraction, reduces overfitting, and enhances the efficient use of spectral information. As a result, the model's classification accuracy and adaptability are significantly improved, making it more effective for HSIC.

E. 3-D–2-D Concatenated CNN

The 3D-CNN [48] is effective in simultaneously capturing both spectral and spatial features, resulting in a more comprehensive feature representation. By integrating these two types of information, the model can extract higher level features. However, the high parameter count associated with 3D-CNNs makes them susceptible to overfitting, particularly when applied to small datasets. To address this challenge, we propose a hybrid approach that concatenates 3D-CNN with 2D-CNN [49], [50], resulting in a deeper network architecture capable of capturing more complex features while effectively managing the parameter count. This integration enhances both model performance and generalization capabilities. Alternatively, a parallel CNN structure can be employed, where separate networks independently learn spectral and spatial features, with their outputs merged at a later stage. This parallel configuration is particularly advantageous for small sample sizes, as it provides a more diverse feature representation and enhances the model's adaptability. In summary, both the hybrid 3D–2D-CNN and parallel CNN strategies effectively address the challenges of small sample sizes and overfitting. They enhance feature extraction and improve the overall performance in HSIC.

IV. Experiments

A. Dataset

In this article, we conducted experiments using three commonly used remote sensing image datasets, namely, Indian Pines (IP), University of Pavia (PU), and Salinas (SA), as shown in Fig. 3. To demonstrate the performance on a small number of training samples, we randomly divided the entire dataset into two nonoverlapping parts, namely, training set and testing set presented in Tables I–III. We only used a limited number of training samples for network training, and the remaining samples were used for performance evaluation.

TABLE I
NUMBER OF SAMPLES OF IP DATASET USED IN THE PROPOSED MODEL

Class No.	Class Name	Total	Training	Validation	Testing
1	Alfalfa	46	1	1	44
2	Corn-notill	1428	15	15	1398
3	Corn-mintill	830	9	9	812
4	Corn	237	3	3	231
5	Grass-pasture	483	5	5	473
6	Grass-trees	730	8	8	714
7	Grass-pasture-mowed	28	1	1	26
8	Hay-windrowed	478	5	5	468
9	Oats	20	1	1	18
10	Soybean-notill	972	10	10	952
11	Soybean-mintill	2455	25	25	2405
12	Soybean-clean	593	6	6	581
13	Wheat	205	3	3	199
14	Woods	1265	13	13	1239
15	Buildings-Grass-Trees-Drives	386	4	4	378
16	Stone-Steel-Towers	93	1	1	91

TABLE II
NUMBER OF SAMPLES OF PU DATASET USED IN THE PROPOSED MODEL

Class No.	Class Name	Total	Training	Validation	Testing
1	Asphalt	6631	67	67	6497
2	Meadows	18649	187	187	18275
3	Gravel	2099	21	21	2057
4	Trees	3064	31	31	3002
5	Painted metal sheets	1345	14	14	1317
6	Bare Soil	5029	51	51	4927
7	Bitumen	1330	14	14	1302
8	Self-Blocking Bricks	3682	37	37	3608
9	Shadows	947	10	10	927

TABLE III
NUMBER OF SAMPLES OF SA DATASET USED IN THE PROPOSED MODEL

Class No.	Class Name	Total	Training	Validation	Testing
1	Brocoli_green_weeds_1	2009	21	21	1967
2	Brocoli_green_weeds_2	3726	38	38	3650
3	Fallow	1976	20	20	1936
4	Fallow_rough_plow	1394	14	14	1366
5	Fallow_smooth	2678	27	27	2624
6	Stubble	3959	40	40	3879
7	Celery	3579	36	36	3507
8	Grapes_untrained	11271	113	113	11045
9	Soil_vinyard_develop	6203	63	63	6077
10	Corn_senesced_green_weeds	3278	33	33	3212
11	Lettuce_romaine_4wk	1068	11	11	1046
12	Lettuce_romaine_5wk	1927	20	20	1887
13	Lettuce_romaine_6wk	916	10	10	896
14	Lettuce_romaine_7wk	1070	11	11	1048
15	Vinyard_untrained	7268	73	73	7122
16	Vinyard_vertical_trellis	1807	19	19	1769

1) The IP dataset contains images with dimensions of 145 × 145 pixels, each with a spatial resolution of 20 m. It includes 220 spectral bands covering a wavelength range from 400 to 2500 nm. After removing 20 bands related to water absorption, the dataset retains 200 spectral bands. The IP dataset includes 16 distinct land cover classes and is annotated with 10 249 samples. These classes represent various types of vegetation, land

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.

use categories, and surface features, with each sample labeled according to its corresponding land cover class.
2) The PU dataset consists of images with a size of 610 × 340 pixels, each with a spatial resolution of 1.3 m. It contains 103 spectral bands with a wavelength range from 430 to 860 nm. The PU dataset covers nine different land cover classes and is labeled with 42 776 samples.
3) The SA dataset comprises images with dimensions of 512 × 217 pixels and a spatial resolution of 3.7 m. It spans a wavelength range from 360 to 2500 nm, consisting of 224 spectral bands. After removing 20 bands related to water absorption, the dataset retains 204 spectral bands. The SA dataset features 16 distinct land cover classes and is annotated with 54 129 samples.

B. Experimental Setup

In our experiments, the hardware and software specifications used were as follows: a 12th-generation Intel Core i7-12700KF CPU, NVIDIA GeForce RTX 3090 GPU 24.0 GB of RAM, and Python 3.9. We set the learning rate to 0.001 and used the cross-entropy function as the loss function. To ensure a fair comparison, we trained the network for 300 epochs. In addition, we chose a fixed spatial window size of 19 × 19.

To reduce spectral dimensionality across different datasets, we applied PCA. Specifically, for the IP and SA datasets, the spectral dimensionality was reduced to 16, while for the PU dataset, it was reduced to 9. During training, the ratio of the training set was fixed at 1%, with each epoch set to 300 and the batch size to 100. We conducted ten training runs and averaged the results to ensure reliability.

The above parameters were set to ensure the fairness and comparability of the experiments. Through these experimental settings, we can evaluate the performance of the proposed method under small sample conditions. These metrics will help assess the efficacy of our method in small sample classification of hyperspectral images.

1) *Overall Accuracy (OA):* It measures the percentage of correctly classified pixels.
2) *Average Accuracy (AA):* It measures the average of OA calculated for each class.
3) *Kappa Coefficient (KAPPA):* A statistical measure used to assess the degree of agreement between qualitative items.

V. Results and Analysis

When comparing models, it is crucial to evaluate the performance and advantages of the proposed model by comparing it with other widely used methods. In this case, several common methods were selected, including SVM [51], 2D-CNN [16], 3D-CNN [17], HybridSN [19], SNMF [21], BSNet-Conv [22], NGNMF-E2DSSA [23], MSFGW [24], BAM [25], CBW [26], SSTTCL2DNN [27], and TACLNN [28]. Since some papers did not provide standard deviation values for the results, the standard deviation, when available, is listed in the footnotes of Tables IV–VI. Bold font indicates the highest accuracy within each category, as well as the best values for OA, AA, and KAPPA.

The detailed descriptions of these comparison models can be found in Section II where we discuss their characteristics and relevance to our proposed approach.

A. Results of IP Dataset

The classification results of different methods, as shown in Fig. 4, along with the accuracy of each class are presented in Table IV. In the IP dataset, different classification methods exhibit varying outcomes. Within the IP dataset, the sample distribution across classes is highly imbalanced, with four classes having fewer than 100 samples each, and the minimum class containing only 20 samples. In such a scenario of limited samples, when the target is set to 1%, only a total of 110 samples are used for training. From the experimental results, the following observations can be made.

1) *Single Model Approaches:* In this limited sample scenario, the OA remains below 65%. For example, the SVM does not adequately utilize the spatial information in hyperspectral images, while the 2D-CNN fails to effectively extract rich spectral information. This results in underutilization of the vast amount of data available in HSI.

2) *3D-CNN Model:* Despite the 3D-CNN model's capability to simultaneously extract spectral and spatial information, it fails to provide sufficient training data and parameter tuning space due to limited samples and thus unable to effectively enhance the accuracy.

3) *Other Models:* While HybridSN combines the strengths of 2D-CNN and 3D-CNN models to increase model complexity, it still performs inadequately on classes with sparse samples. MSFGW used a global weighting mechanism to capture spatial–spectral information at multiple scales, performing relatively better with the highest accuracy in four classes.

4) *MH-2D-3D-CNN Model:* In scenarios where many classes contain fewer than 100 samples, training with only 1% of the data yields a very limited number of training samples. This issue is particularly pronounced in class 6, where the scarcity of samples leads to confusion with class 11. To address these challenges, the MH-2D-3D-CNN model leverages NAMD to enhance key spectral features, while the integration of CBAM with 2D-CNN augments spatial feature extraction. The concatenated architecture of 3D- and 2D-CNN enables simultaneous analysis of both spectral information and spatial information through three distinct pathways, thereby improving feature extraction and overall classification efficiency. The model achieves the highest accuracy in seven classes, with notable performance in classes 1, 2, 3, and 6. By emphasizing critical features through NAMD, the MH-2D-3D-CNN attains superior OA, AA, and Kappa scores, surpassing other methods on the IP dataset.

B. Results of PU Dataset

The classification results of different methods are illustrated in Fig. 5, with the accuracy of each class detailed in Table V. For the PU dataset, the proposed MH-2D-3D-CNN method achieved an OA of 98.13%, an AA of 96.98%, and a Kappa coefficient of 97.19% using a 1% training set.

In this dataset, classes 2, 3, 6, and 8 were relatively more challenging to classify, particularly class 3. This difficulty may

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CLASSIFICATION RESULTS OF THE IP DATASET WITH THE COMPARED METHODS

Class P_{OA}%	SVM [51]	2D-CNN [16]	3D-CNN [17]	HybridSN [19]	SNMF [21]	BSNet-Conv [22]	NGNMF-E2DSSA [23]	MSFGW [24]	MH-2D-3D-CNN
1	18.88	8.89	84.78	31.11	66.78±26.26	81.63±19.08	84.48±17.10	86.93±25.82	**88.89±7.18**
2	46.05	50.74	29.44	50.11	75.37±8.92	81.18±10.69	77.08±7.64	81.71±11.88	**85.30±6.05**
3	45.88	42.14	30.14	46.65	76.05±15.91	82.68±14.86	86.72±7.28	87.25±16.25	**91.39±5.75**
4	30.05	24.79	41.66	20.51	74.63±11.63	64.37±10.05	**78.40±11.41**	67.55±26.68	69.02±12.52
5	71.42	29.50	60.76	50.84	**98.86±2.10**	90.01±1.94	92.73±14.09	96.99±5.11	91.55±3.30
6	74.53	82.83	53.27	81.99	95.34±6.98	94.11±3.55	94.84±3.52	96.83±5.04	**98.49±0.97**
7	25.70	33.33	86.51	40.74	68.84±22.46	73.07±14.52	67.91±24.49	76.20±17.28	**96.67±8.27**
8	87.20	97.67	63.72	**100.0**	98.85±2.63	99.35±1.49	99.50±1.35	99.70±3.83	96.96±8.08
9	18.28	42.11	**95.01**	68.42	44.05±21.53	45.75±23.58	58.69±26.70	88.94±30.95	83.68±14.56
10	50.16	49.48	32.35	81.50	80.68±8.93	82.16±4.77	80.36±9.30	**82.42±6.34**	75.59±2.53
11	52.03	72.84	40.58	74.32	79.37±8.78	83.14±6.58	84.75±5.23	88.00±5.54	**91.69±2.55**
12	34.82	22.83	28.61	47.36	75.98±8.66	77.54±9.22	86.52±7.84	**87.91±7.97**	60.78±7.03
13	76.72	33.66	87.59	98.02	98.43±1.27	96.26±1.55	97.84±2.26	**99.08±3.66**	80.40±5.70
14	79.21	90.97	60.94	93.13	94.08±4.83	91.80±2.74	95.39±3.03	94.01±2.32	**98.23±2.84**
15	48.80	26.96	48.08	30.37	81.59±13.96	76.44±6.80	84.39±14.89	**84.44±5.22**	68.66±6.27
16	**97.50**	19.57	86.79	14.13	95.58±3.90	92.58±3.47	94.30±4.29	95.10±4.43	92.83±9.52
OA	55.98	59.68	69.69	67.52	82.56±6.09	85.45±4.64	85.59±3.72	87.36±6.70	**87.49±0.30**
AA	53.64	45.42	58.67	58.07	81.53±6.09	82.00±5.04	85.25±5.50	**88.32±7.72**	85.79±1.34
KAPPA	48.70	53.30	66.59	63.00	80.00±0.05	83.00±0.05	83.00±0.04	85.00±0.07	**85.70±0.35**

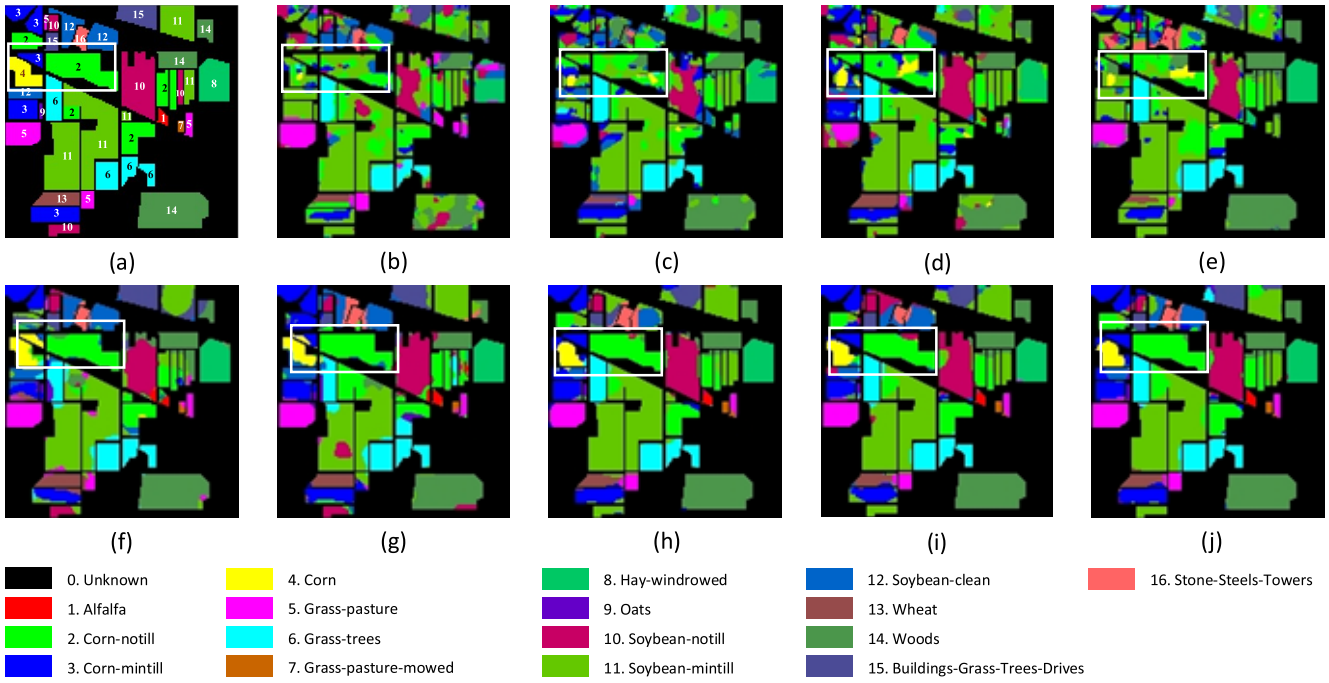


Fig. 4. Land cover categories and classification results of the IP dataset with compared methods: (a) ground truth, (b) SVM, (c) 2D-CNN, (d) 3D-CNN, (e) HybridSN, (f) SNMF, (g) BSNet-Conv, (h) NGNMF-E2DSSA, (i) MSFGW, and (j) MH-2D-3D-CNN.

be due to the significant similarity between classes 1 and 9, making it harder to differentiate between them. Recent classification methods have often struggled to effectively distinguish these two classes. However, the MH-2D-3D-CNN method addressed this issue by using the CBAM to enhance important spatial locations, enabling the 2D-CNN to focus

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.

TABLE V
CLASSIFICATION RESULTS OF THE PU DATASET WITH THE COMPARED METHODS

Class P_{OA}%	SVM [51]	2D-CNN [16]	3D-CNN [17]	HybridSN [19]	BAM [25]	CBW [26]	SSTTCL2DNN [27]	TACLNN [28]	MH-2D-3D-CNN
1	86.99	91.32	89.83	91.78	95.23±0.35	95.32±0.84	88.67±3.68	93.95±0.82	**96.92±2.13**
2	97.33	97.50	96.54	99.77	98.69±1.29	99.05±0.25	97.40±0.53	98.41±0.49	**99.95±0.03**
3	65.38	18.51	70.08	92.24	86.41±1.45	86.84±2.48	71.88±8.56	64.39±6.55	**93.84±3.57**
4	86.22	95.09	95.99	91.01	**98.47**±0.27	98.44±0.85	83.16±2.26	92.19±2.45	94.85±2.86
5	98.39	99.19	99.72	97.76	99.52±0.32	**99.88**±0.15	88.84±8.12	96.07±3.26	99.84±0.04
6	76.48	89.59	80.84	99.38	98.36±1.12	**99.84**±0.11	80.10±9.41	93.87±4.29	99.57±0.19
7	78.58	42.90	69.64	**96.83**	85.78±1.46	88.88±0.62	72.67±5.18	73.45±5.82	96.27±1.88
8	85.33	87.04	80.31	90.72	87.91±1.42	90.92±1.36	85.27±2.38	88.95±2.01	**97.58±1.34**
9	96.09	**97.75**	96.70	72.17	95.85±2.11	96.60±1.74	76.05±13.78	80.45±3.02	94.02±2.43
OA	89.19	89.02	90.19	95.94	95.76±0.36	95.83±0.31	89.18±0.66	93.01±0.70	**98.13±0.38**
AA	85.64	79.87	86.63	92.41	94.55±1.08	95.09±0.93	82.67±1.67	86.86±0.72	**96.98±1.61**
KAPPA	85.66	85.42	87.01	94.95	94.66±0.34	94.77±0.27	85.55±0.96	90.72±0.95	**97.19±0.11**

TABLE VI
CLASSIFICATION RESULTS OF THE SA DATASET WITH THE COMPARED METHODS

Class P_{OA}%	SVM [51]	2D-CNN [16]	3D-CNN [17]	HybridSN [19]	BAM [25]	CBW [26]	SSTTCL2DNN [27]	TACLNN [28]	MH-2D-3D-CNN
1	99.28	59.97	97.54	99.72	99.97±1.26	**99.98**±1.05	94.27±4.58	98.12±2.12	99.59±0.12
2	99.24	99.48	98.89	99.92	99.87±0.75	99.88±0.88	96.82±0.89	98.76±0.65	**99.97±0.01**
3	77.03	60.01	97.42	98.68	98.28±0.92	99.45±0.62	99.10±0.79	98.93±0.80	**99.59±0.05**
4	96.52	98.27	98.10	97.19	99.86±0.59	96.28±0.26	98.21±1.88	98.36±1.88	**100.0±0.00**
5	96.13	94.80	97.78	**99.39**	98.79±1.37	97.81±0.38	98.98±0.26	98.79±1.64	96.34±0.67
6	99.60	99.89	97.97	99.85	99.95±1.42	99.97±1.02	**99.98±0.03**	99.46±0.91	99.97±0.02
7	99.55	97.21	98.71	99.99	99.93±0.48	98.11±2.01	97.97±1.61	99.76±0.21	**100.0±0.00**
8	89.81	83.53	89.67	94.84	88.29±3.42	90.99±1.78	89.15±0.17	95.52±0.92	**99.03±0.36**
9	99.83	99.26	98.99	99.99	99.82±0.94	99.59±0.27	99.64±0.57	99.73±0.29	**100.0±0.00**
10	93.09	84.95	96.27	96.70	97.63±0.38	98.03±0.68	97.90±1.95	**98.98±1.10**	98.55±1.09
11	75.83	90.00	94.48	99.46	97.00±0.92	97.53±1.59	97.82±0.62	94.26±2.08	**100.0±0.00**
12	92.86	97.15	98.76	99.79	99.93±0.84	99.82±1.23	99.55±0.42	99.42±0.87	**100.0±0.00**
13	89.16	95.74	95.88	98.74	98.34±0.72	**99.98**±0.24	97.61±1.79	97.87±2.00	98.89±0.33
14	86.36	94.84	98.94	99.03	98.66±1.64	**99.35**±1.18	99.24±0.59	99.02±1.69	98.48±1.14
15	54.64	72.57	86.18	91.69	78.18±4.68	87.80±3.87	94.14±1.08	97.10±2.30	**99.83±0.02**
16	95.58	91.12	98.70	99.52	**99.83**±0.21	96.22±4.57	90.20±5.79	99.12±1.47	96.03±1.77
OA	88.76	87.61	95.24	97.35	95.49±1.26	95.82±0.31	95.65±0.32	98.02±0.09	**99.22±0.21**
AA	90.28	88.66	96.78	97.91	97.10±0.59	97.55±0.26	96.91±0.56	98.33±0.27	**99.14±0.35**
KAPPA	87.43	86.22	94.66	96.47	94.75±1.16	95.27±0.30	95.16±0.36	97.79±0.10	**99.13±0.12**

on critical features. For class 6, which involves line-like structures in the image, the model employed NAMD to strengthen spectral features and separate the background from other areas more effectively. Overall, MH-2D-3D-CNN

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.

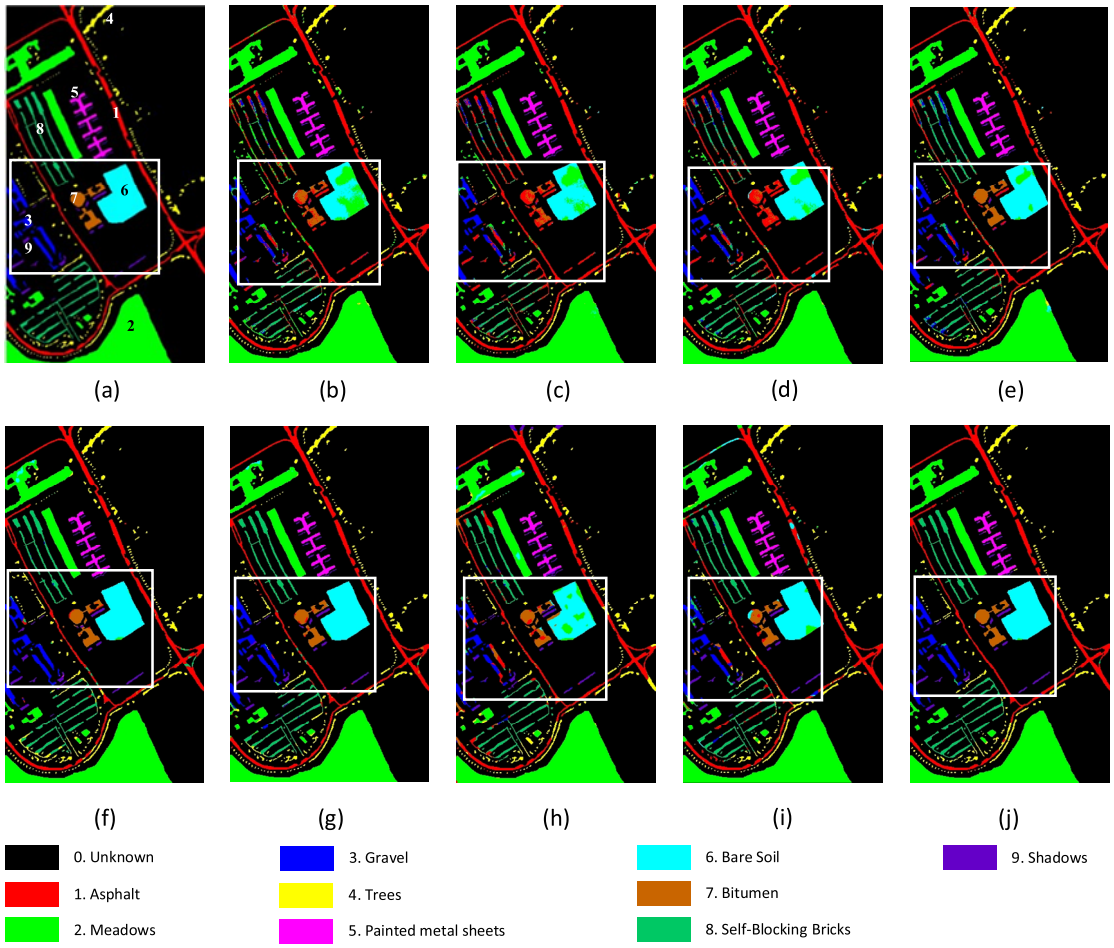


Fig. 5. Land cover categories and classification results of the PU dataset with compared methods: (a) ground truth, (b) SVM, (c) 2D-CNN, (d) 3D-CNN, (e) HybridSN, (f) BAM, (g) CBW, (h) SSTTCL2DNN, (i) TACLNN, and (j) MH-2D-3D-CNN.

achieved the best classification results for four out of nine categories, outperforming other methods in terms of OA. This indicates that MH-2D-3D-CNN exhibits higher recognition rates compared to alternative methods, resulting in superior overall performance.

C. Results of SA Dataset

The classification results of various methods are illustrated in Fig. 6, with the accuracy of each class summarized in Table VI. In the SA dataset, the total number of samples is substantial, with each class containing at least 900 samples. Even when utilizing only 1% of the data for training, the MH-2D-3D-CNN model benefited from 549 labeled samples, yielding OA, AA, and Kappa values of 99.22%, 99.14%, and 99.13%, respectively. Notably, classes 8 and 15 presented significant challenges in differentiation within the SA dataset.

The performance comparison revealed that the 3D-CNN outperformed the 2D-CNN, while the HybridSN demonstrated superior results compared to the 3D-CNN. To address these insights, the MH-2D-3D-CNN model was developed by concatenating 3D-CNN and 2D-CNN architectures for more effective feature extraction. In addition, the incorporation of the CBAM into the 2D-CNN enhanced the model's performance by reinforcing spatial features while leveraging the broader spatial–spectral features captured by the 3D-CNN. This combination allowed the MH-2D-3D-CNN model to exceed the performance of the HybridSN's hybrid structure.

Consequently, the model achieved a remarkable accuracy of 100% in five classes, showcasing superior recognition rates compared to other methods such as SSTTCL2DNN [27] and TACLNN [28]. Thus, the MH-2D-3D-CNN model demonstrated outstanding OA, distinguishing itself from alternative approaches.

VI. Discussion

Based on the results presented in Tables IV–VI, the following conclusions can be drawn. The proposed MH-2D-3D-CNN method achieves the highest OA, AA, and KAPPA values across different datasets. By enhancing and combining features from all three channels for classification, MH-2D-3D-CNN achieves near-optimal accuracy in each class. In the IP dataset, due to the lower number of samples in classes 1, 7, 9, and 16, as well as the high similarity between classes 7 and 9, the OA is slightly higher than the MSFGW model's 0.13%.

To verify the feasibility of the proposed MH-2D-3D-CNN method, we compared it with traditional statistical methods and other CNN models proposed in recent years, including SVM, 2D-CNN, 3D-CNN, HybridSN, SNMF, BSNet-Conv, NGNMF-E2DSSA, MSFGW, BAM, CBW, SSTTCL2DNN, and TACLNN. The tables illustrate the imbalance in the datasets, with significant differences in accuracy ranging from

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.

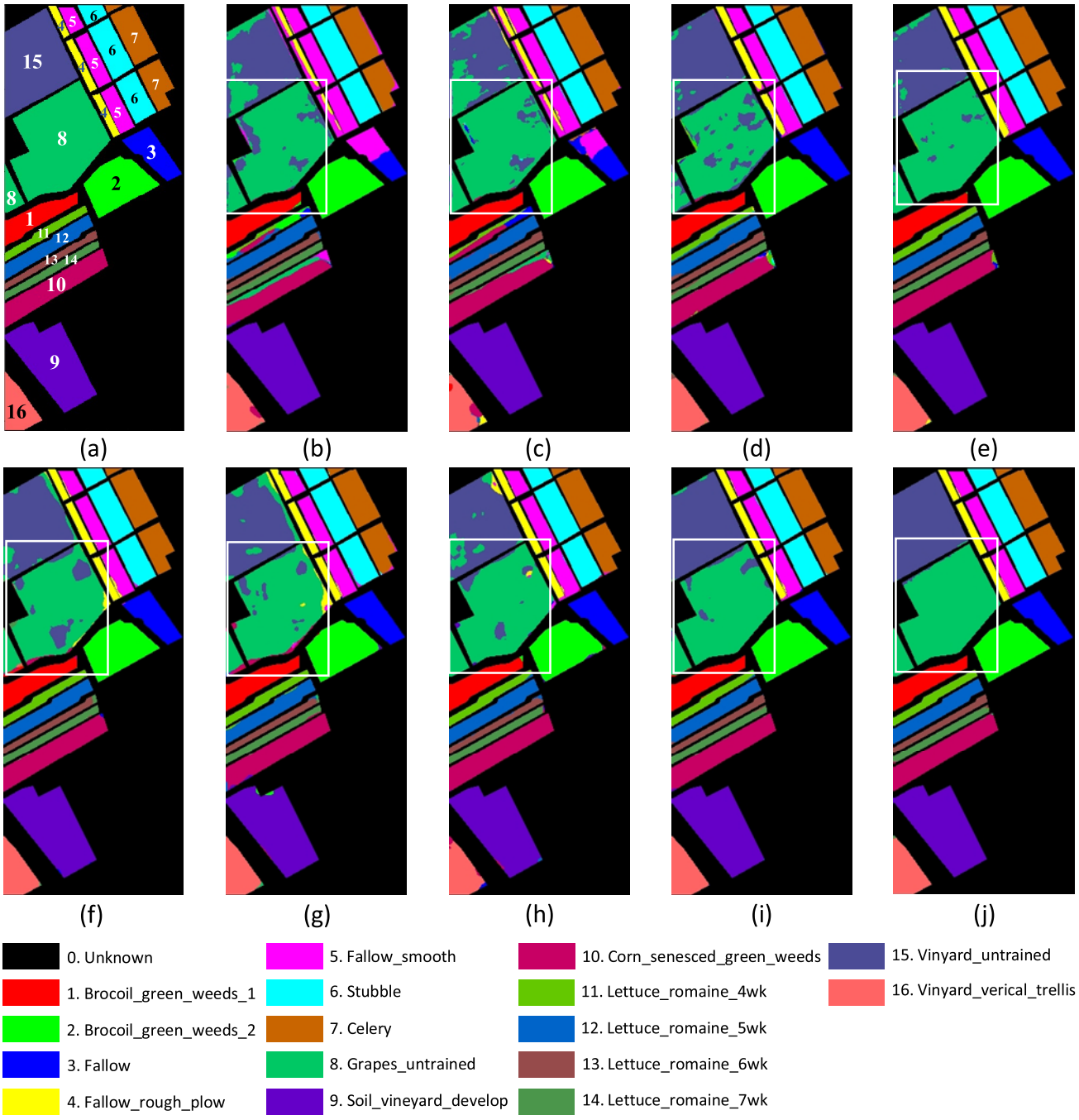


Fig. 6. Land cover categories and classification results of the SA dataset with compared methods: (a) ground truth, (b) SVM, (c) 2D-CNN, (d) 3D-CNN, (e) HybridSN, (f) BAM, (g) CBW, (h) SSTTCL2DNN, (i) TACLNN, and (j) MH-2D-3D-CNN.

approximately 1%–30% across various models. It is noteworthy that the proposed MH-2D-3D-CNN method achieves the highest accuracy across all datasets, outperforming other models.

A. Different Window Size

This article conducted a comparison of different sizes of adjacent pixels, including 15, 17, 19, 21, and 23, using the MH-2D-3D-CNN on the IP, PU, and SA datasets. As shown in Table VII, the results indicated that varying window sizes had different impacts on the OA of the model. Specifically, a window size of 19 achieved the highest OA, with the accuracy gradually decreasing as the window size increased beyond 19. This suggests that a window size of 19 provided the best performance in this experiment. Based on these results, we fixed the window size at 19 for all subsequent experiments to ensure optimal results.

B. Ablation Analysis

In this section, we explore the impact of each channel on the experimental results. Our model was trained on datasets such as IP, PU, and SA divided into three channels for evaluation: Channel-1 utilizing NAMD combined with 2D-CNN, Channel-2 employing CBAM combined with 2D-CNN, and Channel-3 utilizing a concatenated 3D–2D-CNN approach. We conducted ten training iterations and averaged the results,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
CLASSIFICATION PERFORMANCE ON DIFFERENT WINDOW SIZES

Window Size	Indian Pines	University of Pavia	Salinas
15×15	80.67±1.28	97.70±0.16	98.01±0.14
17×17	86.01±2.11	96.23±0.07	98.21±0.31
19×19	**87.49±0.30**	**98.13±0.38**	**99.22±0.21**
21×21	84.53±1.37	97.42±0.12	99.08±1.45
23×23	83.74±2.07	96.79±0.13	96.01±0.11

with a specific focus on analyzing the concepts proposed by each channel.

In the experiments, we compared the performance of single-channel and multichannel fusion configurations of Channel-1 with PCA and Channel-1 with NAMD. In the single-channel setup, Channel-1 with PCA demonstrated stronger performance, particularly in scenarios with small sample sizes. PCA's dimensionality reduction effectively reduces noise and emphasizes key spectral features, making it highly suitable for hyperspectral data classification under limited sample conditions. However, in the multichannel fusion configuration, Channel-1 with NAMD outperformed the PCA-based approach. NAMD's capacity for enhancing spectral target detection proved more beneficial when integrated with spatial features from the other channels, as it introduced critical spectral information that PCA might overlook. This comparison is explored further in Section VI-D.

As shown in Table VIII, Channel-1 exhibited the weakest individual performance, while Channel-3 surpassed both Channel-1 and Channel-2, excelling in feature extraction and delivering stronger overall results. The combination of all three channels effectively mitigated their individual limitations, yielding the best classification outcomes.

Although Channel-1 on its own performed the worst, combining Channel-1 with Channel-3 and Channel-2 with Channel-3 improved accuracy compared to Channel-2 alone, underscoring Channel-3's effectiveness in capturing spectral–spatial information. While Channel-1 extracts distinct spectral features that are crucial when working with small sample sizes, spatial information remains vital when more data is available.

Overall, combining all three channels significantly enhances performance, proving the model's effectiveness in HSIC.

1) Indian Pines: In Table VIII, the IP dataset poses significant challenges due to sample imbalance and sparse training data, making it difficult to enhance training effectiveness. Upon analysis, a notable gap is observed in the performance of Channel-3 during training, compared to other methods. However, the addition of Channel-1 and Channel-2, individually, results in a modest improvement in accuracy. This suggests that, despite the lower individual performance of Channel-1 and Channel-2, they complement Channel-3 effectively, contributing to the overall performance enhancement.

The accuracy of Channel-1 is slightly lower than that of Channel-2, indicating that emphasizing spatial relationships, as Channel-2 does, is more impactful than simply increasing data diversity and reinforcing spectral features, as Channel-1 achieves. Nevertheless, the integration of all three channels leads to a further improvement in accuracy, underscoring the superiority of the MH-2D-3D-CNN model on the IP dataset.

The key advantage of the MH-2D-3D-CNN model lies in its ability to compensate for the limitations of each individual channel, thereby improving overall classification performance. This integrative approach proves highly effective in addressing the complexities of the IP dataset, resulting in the best classification outcomes.

2) University of Pavia: In the PU dataset, the MH-2D-3D-CNN model exhibits outstanding performance, achieving an OA of 98.13%, an AA of 96.98%, and a Kappa coefficient of 97.19%. The following trends were observed: Channel-3 consistently outperforms the other two channels in terms of accuracy. While combining Channel-2 with Channel-1 results in a slight performance decrease, the addition of both to Channel-3 significantly enhances OA. Notably, the combination of Channel-1 and Channel-3 yields better results than the pairing of Channel-2 and Channel-3, suggesting that enhancing spectral–spatial feature extraction (the strength of Channel-3) is more effective than focusing solely on spatial signal enhancement (the focus of Channel-2).

Ultimately, the integration of all three channels allows the MH-2D-3D-CNN to achieve the highest OA, highlighting the value of multichannel combinations in boosting model performance. This demonstrates the model's superiority in the PU dataset classification and validates the effectiveness of incorporating diverse feature extraction strategies for improving classification outcomes.

3) Salinas: In the SA dataset, classification results are influenced by the relatively large number of samples, with even the least populated class containing 900 samples, making classification with a 1% training set more manageable. The following trends were observed; Channel-3 consistently achieves higher accuracy compared to the other two channels. However, when Channel-3 is combined with either Channel-1 or Channel-2, a slight decrease in performance is noted. This reduction is likely due to the similarity in the results from each individual channel, which interferes with Channel-3's optimal feature extraction, diminishing its effectiveness. Conversely, combining Channel-1 and Channel-2 results in a marginal performance improvement, and when all three channels are integrated, the overall performance is further enhanced.

Ultimately, the MH-2D-3D-CNN model, through the combination of all three channels, achieves the highest OA. This reinforces the importance of multichannel integration in enhancing model performance, as the MH-2D-3D-CNN delivers superior classification results on the SA dataset.

C. Different Numbers of Training Samples

To further validate the performance of the MH-2D-3D-CNN model, we addressed the challenge of limited sample sizes in certain classes of the IP dataset, which made it impractical to reduce the training set to 0.5%. Instead, we chose to evaluate the model by reducing the training data to 0.5% using the PU and SA datasets, ensuring a robust assessment of the model's effectiveness under conditions of extreme sample scarcity.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
CLASSIFICATION RESULTS OF IP, PU, AND SA DATASETS WITH THE COMPARED METHODS

METHOD	Indian Pines			University of Pavia			Salinas		
	OA	AA	KAPPA	OA	AA	KAPPA	OA	AA	KAPPA
Channel-1(PCA)	80.42±2.30	76.44±2.23	77.96 ±2.60	95.58±0.23	93.78±0.57	94.79±0.31	96.87±0.40	96.73±0.42	96.75±0.45
Channel-1 (NAMD)	72.88±3.21	63.52±5.44	60.20±3.91	88.68±1.76	80.41±2.34	84.96±1.30	96.00±0.25	95.60±0.28	95.55±0.17
Channel-2	82.64±1.29	74.90±2.05	79.90±1.43	96.41±0.84	93.06±1.88	95.41±0.14	98.27±0.31	97.91±0.21	98.19±0.12
Channel-3	85.29±2.19	82.53±3.28	82.71±2.47	97.45±0.77	95.48±1.31	96.62±0.34	99.12±0.22	98.86±0.25	99.02±0.05
Channel-1+2	83.63±1.20	79.53±2.00	80.46±1.36	96.23±0.95	94.13±1.57	95.00±0.22	98.65±0.71	98.08±0.33	98.50±0.11
Channel-1+3	85.55±1.08	82.40±1.43	83.03±1.24	97.66±0.33	95.21±1.28	96.90±0.40	98.23±0.25	98.30±0.37	98.03±0.18
Channel-2+3	85.73±1.63	84.15±1.41	83.47±1.83	97.57±0.53	95.38±1.64	95.38±0.15	98.14±0.15	98.02±0.43	98.04±0.20
Channel-1(PCA)+2+3	86.03±1.71	84.79±1.96	84.08±1.90	97.71±2.63	95.91±3.65	96.97±2.63	98.69±0.22	97.82±0.41	98.55±0.27
MH-2D-3D-CNN	**87.49±0.30**	**85.79±1.34**	**85.70±0.35**	**98.13±0.38**	**96.98±1.61**	**97.19±0.11**	**99.22±0.21**	**99.14±0.35**	**99.13±0.12**

TABLE IX
CLASSIFICATION RESULTS OF THE PU DATASET WITH THE COMPARED METHODS (0.5%)

METHOD	1%			0.5%		
	OA	AA	KAPPA	OA	AA	KAPPA
SVM [51]	89.19	85.64	85.66	78.53	72.63	83.38
2D-CNN [16]	89.02	79.87	85.42	80.37	76.59	76.98
3D-CNN [17]	90.19	86.63	87.01	81.95	75.21	68.25
3D-2D CNN [18]	96.96±0.14	95.11±0.19	96.24±0.11	90.50	84.08	88.44
HybridSN [19]	94.44±0.32	88.23±0.14	92.52±0.39	89.12	80.76	86.26
F3D [52]	95.18±0.27	90.79±0.71	93.66±0.28	90.36	88.63	87.97
MSDAN [53]	96.14±0.19	94.91±0.26	94.74±0.19	93.31	88.89	91.02
GloRe-HCNN [29]	97.48±0.21	95.65±0.34	96.65±0.25	94.40	90.01	90.50
VIT [33]	-	-	-	91.27	88.19	88.39
miniGCN [54]	-	-	-	92.42	84.80	89.95
SSFTT [55]	-	-	-	93.89	91.42	91.85
MH-2D-3D-CNN	**98.13±0.38**	**96.98±1.61**	**97.19±0.11**	**94.48±0.71**	**92.53±0.30**	**91.32±0.26**

For the PU dataset, as shown in Table IX, there were still 216 labeled samples available for training with the 0.5% training set. MH-2D-3D-CNN outperformed other methodologies, including F3-D [52], MSDAN [53], GloRe-HCNN [29], ViT [33], miniGCN [54], and SSFTT [55] achieving OA, AA, and KAPPA values of 94.48%, 92.53%, and 91.32%, respectively.

In the SA dataset, as shown in Table X, the MH-2D-3D-CNN model achieved a significantly higher OA compared to the SSFTT model, surpassing it by 1.03%. In addition, when the training data were reduced to 0.5%, the performance of other methods exhibited a sharp decline, as they were unable to capture enough data to extract relevant features for each class. Although the accuracy of the MH-2D-3D-CNN model also experienced a slight decrease under this reduced training scenario, it continued to outperform other methods. These findings demonstrate that the MH-2D-3D-CNN model offers more accurate and robust HSIC, even under conditions of SLTSS, thereby maintaining a superior level of performance relative to competing approaches.

D. Comparative Analysis of NAMD and PCA in Channel-1

This section provides a detailed comparison between NAMD and PCA when applied to Channel-1, with results evaluated across the IP, PU, and SA datasets. Key performance metrics, including OA, AA, and Kappa, are summarized in Table VIII. While Channel-1 with PCA consistently demonstrates better performance across all datasets, NAMD proves to be more effective in the multichannel fusion configuration, as evidenced in the same table.

NAMD, specifically designed for target detection, isolates and emphasizes critical spectral information by leveraging statistical properties. As demonstrated in [15], this method has been particularly effective in HSIC. When integrated with the other branches in the MH-2D-3D-CNN model, NAMD

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.

TABLE X
CLASSIFICATION RESULTS OF THE SA DATASET WITH THE COMPARED METHODS (0.5%)

METHOD	1%			0.5%		
	OA	AA	KAPPA	OA	AA	KAPPA
SVM [51]	88.76	90.28	87.43	77.28	76.51	82.88
2D-CNN [16]	87.61	88.66	86.22	83.37	81.40	86.96
3D-CNN [17]	95.24	96.78	94.66	84.02	81.98	87.02
3D-2D CNN [18]	96.63±0.17	96.96±0.14	95.88±1.24	90.34	89.58	91.98
HybridSN [19]	97.35±0.34	97.91±1.13	96.47±1.67	94.07	93.41	93.66
VIT [33]	-	-	-	95.37	94.36	94.84
miniGCN [54]	-	-	-	94.47	92.75	93.85
SSFTT [55]	-	-	-	96.35	95.10	95.94
MH-2D-3D-CNN	**99.22±0.21**	**99.14±0.35**	**99.13±0.12**	**97.38±0.55**	**97.09±0.24**	**97.23±0.28**

TABLE XI
TRAINING TIME OF DIFFERENT ALGORITHMS

Time (Seconds)	Channel-1+2	Channel-1+3	Channel-2+3	MH2D-3D-CNN
Indian Pines	24.114	25.827	36.300	38.607
University of Pavia	80.244	50.204	97.632	106.333
Salinas	105.143	110.584	175.498	182.581

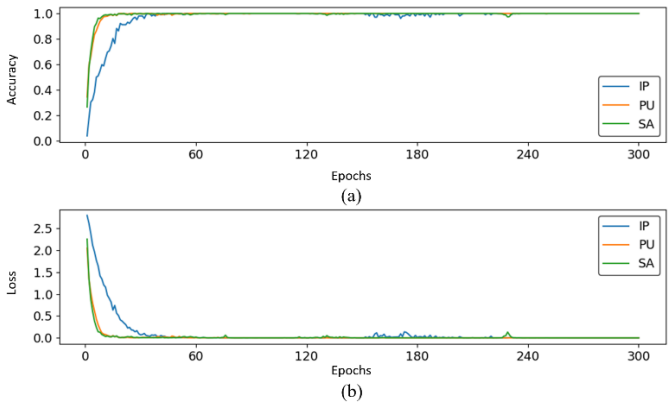


Fig. 7. (a) Training accuracy and (b) training loss of the MH-2D-3D-CNN.

contributes additional feature information and dimensionality reduction, enhancing the overall performance of the network.

In contrast, PCA primarily focuses on dimensionality reduction by transforming data into orthogonal components. Although PCA performs well when applied to individual channels, its repeated use in Channel-1 offers limited benefit in a multichannel setup. This is because the dimensionality reduction process overlaps with feature extraction from other branches, which diminishes the additional contributions PCA might offer in a fusion-based model.

Conversely, NAMD enriches the feature space by introducing target-specific spectral information, which complements the spatial and attention-based features extracted by the other channels. This is reflected in Table VIII, where the MH-2D-3D-CNN, incorporating NAMD in Channel-1, achieves superior results, surpassing PCA in terms of OA. NAMD's key advantage lies in its ability to preserve and emphasize critical spectral features, which allows for better class differentiation and more effective dimensionality reduction in HSI.

By focusing on distinct, target-specific spectral characteristics, NAMD introduces unique spectral features to capture subtle class distinctions that may be overlooked by PCA. Consequently, NAMD emerges as the more suitable choice for Channel-1, playing a pivotal role in enhancing HSIC accuracy and enriching the model's overall feature space when integrated with other branches of the network. By incorporating complementary spectral information, NAMD strengthens the model's capacity to differentiate between classes, ultimately resulting in improved classification performance.

E. Convergence and Complexity Analysis

This section analyzes Fig. 7(a) and (b), which depict the accuracy and loss of the IP, PU, and SA datasets during training for 300 epochs. The vertical axis represents accuracy and loss, while the horizontal axis represents the number of epochs. All three datasets stabilize around 200 epochs. Table XI compares the training times of the three datasets under different training models. It can be observed from the table that the MH-2D-3D-CNN model takes the longest time due to its higher number of channels, but it also achieves the best accuracy performance. Table XI reveals that the training times correspond to the dataset sizes, with SA having the largest number of samples and IP having the fewest.

VII. CONCLUSION

The challenge of effectively extracting features with an SLTSS remains a significant obstacle in HSIC. This article presents an innovative MH-2D-3D-CNN model specifically designed to address this issue. The model is organized into three distinct channels, each dedicated to extracting specific spectral and spatial features from hyperspectral images.

1) *First Channel:* It employs the NAMD to enhance spectral features, followed by a 2D-CNN to effectively capture spatial features.
2) *Second Channel:* It integrates 2D-CNN with the CBAM to highlight important spatial details while suppressing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.

irrelevant information, thereby improving classification accuracy.

3) *Third Channel:* It combines a concatenated architecture of 3D-CNN and 2D-CNN to comprehensively capture both spectral and spatial features.

The MH-2D-3D-CNN model enhances feature extraction by integrating target detection techniques and attention mechanisms. These mechanisms operate in parallel across the three channels, enabling independent and efficient training processes. This approach significantly improves overall model performance in HSIC tasks, especially under SLTSS conditions. Experimental results on three publicly available HIS datasets (IP, PU, and SA) using 1% and 0.5% labeled training samples highlight the significant performance advantages of the proposed MH-2D-3D-CNN model. The findings demonstrate the model's superiority over state-of-the-art methods, particularly in HSIC tasks with SLTSS, showcasing its robustness and accuracy in challenging scenarios.

Future work may encompass refining the model architecture, investigating advanced data augmentation techniques, implementing transfer learning, assessing practical applicability, and enhancing model robustness and interpretability.

REFERENCES

[1] B. Rasti et al., "Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 60–88, Dec. 2020.

[2] J. W. P. Sun, H. Li, W. Li, X. Meng, C. Ge, and Q. Du, "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 10–43, Jun. 2021.

[3] N. Yokoya, J. Chan, and K. Segl, "Potential of resolution-enhanced hyperspectral data for mineral mapping using simulated EnMAP and Sentinel-2 images," *Remote Sens.*, vol. 8, no. 3, p. 172, Feb. 2016.

[4] I. C. C. Acosta, M. Khodadadzadeh, L. Tusa, P. Ghamisi, and R. Gloaguen, "A machine learning framework for drill-core mineral mapping using hyperspectral and high-resolution mineralogical data fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4829–4842, Dec. 2019.

[5] X. Yang and Y. Yu, "Estimating soil salinity under various moisture conditions: An experimental study," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2525–2533, May 2017.

[6] Y. Han, J. Li, Y. Zhang, Z. Hong, and J. Wang, "Sea ice detection based on an improved similarity measurement method using hyperspectral data," *Sensors*, vol. 17, no. 5, p. 1124, May 2017.

[7] S.-Y. Chen, Y.-C. Cheng, W.-L. Yang, and M.-Y. Wang, "Surface defect detection of wet-blue leather using hyperspectral imaging," *IEEE Access*, vol. 9, pp. 127685–127702, 2021.

[8] S.-Y. Chen, M.-Y. Wang, Y.-M. Kuo, Y.-C. Chan, and Y.-C. Chen, "Almond defect and freshness inspection system using hyperspectral imaging and deep learning techniques," *Postharvest Biol. Technol.*, vol. 211, May 2024, Art. no. 112837.

[9] S.-Y. Chen, M.-F. Chiu, and X.-W. Zou, "Real-time defect inspection of green coffee beans using NIR snapshot hyperspectral imaging," *Comput. Electron. Agricult.*, vol. 197, Jun. 2022, Art. no. 106970, doi: 10.1016/j.compag.2022.106970.

[10] S.-Y. Chen, C.-Y. Chang, C.-S. Ou, and C.-T. Lien, "Detection of insect damage in green coffee beans using VIS-NIR hyperspectral imaging," *Remote Sens.*, vol. 12, no. 15, p. 2348, Jul. 2020.

[11] S.-Y. Chen, S.-H. Hsu, C.-Y. Ko, and K.-H. Hsu, "Real-time defect and freshness inspection on chicken eggs using hyperspectral imaging," *Food Control*, vol. 11, Aug. 2023, Art. no. 109716.

[12] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[13] J. Xia, L. Bombrun, Y. Berthoumieu, C. Germain, and P. Du, "Spectral–spatial rotation forest for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2016, pp. 5126–5129.

[14] E. Blanzieri and F. Melgani, "Nearest neighbor classification of remote sensing images with the maximal margin principle," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 6, pp. 1804–1811, Jun. 2008.

[15] S.-Y. Chen, K.-H. Hsu, and T.-H. Kuo, "Hyperspectral target detection-based 2-D–3-D parallel convolutional neural networks for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9451–9469, 2024, doi: 10.1109/JSTARS.2024.3394704.

[16] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[17] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[18] S. Ghaderizadeh, D. Abbasi-Moghadam, A. Sharifi, N. Zhao, and A. Tariq, "Hyperspectral image classification using a hybrid 3D-2D convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7570–7588, 2021.

[19] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D-2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[20] Y. Gu, Q. Wang, H. Wang, D. You, and Y. Zhang, "Multiple kernel learning via low-rank nonnegative matrix factorization for classification of hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2739–2751, Jun. 2015.

[21] J.-M. Li and Y.-T. Qian, "Clustering-based hyperspectral band selection using sparse nonnegative matrix factorization," *J. Zhejiang Univ. Sci. C*, vol. 12, no. 7, pp. 542–549, Jul. 2011.

[22] Y. Cai, X. Liu, and Z. Cai, "BS-Nets: An end-to-end framework for band selection of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1969–1984, Mar. 2020.

[23] H. Fu et al., "A novel band selection and spatial noise reduction method for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535713.

[24] J. Wang, J. Liu, J. Cui, J. Luan, and Y. Fu, "Multiscale fusion network based on global weighting for hyperspectral feature selection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 2977–2991, 2023.

[25] H. Dong, L. Zhang, and B. Zou, "Band attention convolutional networks for hyperspectral image classification," 2019, *arXiv:1906.04379*.

[26] L. Zhao, J. Yi, X. Li, W. Hu, J. Wu, and G. Zhang, "Compact band weighting module based on attention-driven for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9540–9552, Nov. 2021.

[27] W.-S. Hu, H.-C. Li, T.-Y. Ma, Q. Du, A. Plaza, and W. J. Emery, "Hyperspectral image classification based on tensor-train convolutional long short-term memory," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 858–861.

[28] W.-S. Hu, H.-C. Li, Y.-J. Deng, X. Sun, Q. Du, and A. Plaza, "Lightweight tensor attention-driven ConvLSTM neural network for hyperspectral image classification," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 3, pp. 734–745, Apr. 2021.

[29] W. Wang, X. Ma, L. Leng, Y. Wang, B. Liu, and J. Sun, "A hybrid CNN based on global reasoning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2022.3199208.

[30] Z. Wang, C. Zou, and W. Cai, "Small sample classification of hyperspectral remote sensing images based on sequential joint deeping learning model," *IEEE Access*, vol. 8, pp. 71353–71363, 2020, doi: 10.1109/ACCESS.2020.2986267.

[31] Z. Li et al., "Deep multilayer fusion dense network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1258–1270, 2020, doi: 10.1109/JSTARS.2020.2982614.

[32] J. Xiang, C. Wei, M. Wang, and L. Teng, "End-to-end multilevel hybrid attention framework for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3126125.

[33] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.

[34] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615, doi: 10.1109/TGRS.2021.3130716.
[35] Z. Zheng, Y. Zhong, A. Ma, and L. Zhang, "FPGA: Fast patch-free global learning framework for fully end-to-end hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5612–5626, Aug. 2020, doi: 10.1109/TGRS.2020.2967821.
[36] R. Ji, K. Tan, X. Wang, C. Pan, and L. Xin, "PASSNet: A spatial–spectral feature extraction network with patch attention module for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023, doi: 10.1109/LGRS.2023.3322422.
[37] X. Wang, K. Tan, P. Du, C. Pan, and J. Ding, "A unified multiscale learning framework for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4508319, doi: 10.1109/TGRS.2022.3147198.
[38] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014, doi: 10.1109/TGRS.2022.3207933.
[39] C.-I. Chang, "Hyperspectral anomaly detection: A dual theory of hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511720, doi: 10.1109/TGRS.2021.3086768.
[40] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 3–19.
[41] D. Wei, P. Li, H. Xie, and Y. Xu, "DRCNet: Road extraction from remote sensing images using DenseNet with recurrent criss-cross attention and convolutional block attention module," *IEEE Access*, vol. 11, pp. 126879–126891, 2023.
[42] S. Koffas, S. Picek, and M. Conti, "Dynamic backdoors with global average pooling," in *Proc. IEEE 4th Int. Conf. Artif. Intell. Circuits Syst. (AICAS)*, Incheon, South Korea, Jun. 2022, pp. 320–323, doi: 10.1109/AICAS54282.2022.9869920.
[43] S. Meng, L.-T. Huang, and W.-Q. Wang, "Tensor decomposition and PCA jointed algorithm for hyperspectral image denoising," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 7, pp. 897–901, Jul. 2016, doi: 10.1109/LGRS.2016.2552403.
[44] M. Mottus, M. Molinier, E. Halme, H. Cu, and J. Laaksonen, "Patch size selection for analysis of sub-meter resolution hyperspectral imagery of forests," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Brussels, Belgium, Jul. 2021, pp. 2035–2038, doi: 10.1109/IGARSS47720.2021.9554257.
[45] T. Kumar Dutta and D. Ranjan Nayak, "CDANet: Channel split dual attention based CNN for brain tumor classification in mr images," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Bordeaux, France, Oct. 2022, pp. 4208–4212, doi: 10.1109/ICIP46576.2022.9897799.
[46] X. Yang, C. Yu, J. Gao, Y. Wang, and H. Yang, "SAVE: Spatial-attention visual exploration," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Bordeaux, France, Oct. 2022, pp. 1356–1360, doi: 10.1109/ICIP46576.2022.9897714.
[47] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778, doi: 10.1109/CVPR.2016.90.
[48] M. Zhao, S. Shi, J. Chen, and N. Dobigeon, "A 3-D-CNN framework for hyperspectral unmixing with spectral variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521914, doi: 10.1109/TGRS.2022.3141387.
[49] H. Guo, J. Liu, J. Yang, Z. Xiao, and Z. Wu, "Deep collaborative attention network for hyperspectral image classification by combining 2-D CNN and 3-D CNN," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4789–4802, 2020, doi: 10.1109/JSTARS.2020.3016739.
[50] C. Yu, R. Han, M. Song, C. Liu, and C.-I. Chang, "A simplified 2D-3D CNN architecture for hyperspectral image classification based on spatial–spectral fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2485–2501, 2020, doi: 10.1109/JSTARS.2020.2983224.
[51] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, Sep. 1995.
[52] M. Ahmad, A. M. Khan, M. Mazzara, S. Distefano, M. Ali, and M. S. Sarfraz, "A fast and compact 3-D CNN for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[53] X. Wang and Y. Fan, "Multiscale densely connected attention network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1617–1628, 2022.
[54] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.
[55] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral-spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.



Shih-Yu Chen (Member, IEEE) received the B.S. degree in electrical engineering from Da-Yeh University, Dacun, Taiwan, in 2005, the M.S. degree in electrical engineering from National Chung Hsing University, Taichung, Taiwan, in 2010, and the Ph.D. degree in electrical engineering from the University of Maryland at Baltimore County (UMBC), Baltimore, MD, USA, in 2014.

Since 2018, he has been with the National Yunlin University of Science and Technology, Douliu, Yunlin, Taiwan, where he is currently an Associate Professor with the Department of Computer Science and Information Engineering. His research interests include multi/hyperspectral image processing, medical signal processing, image processing, and pattern recognition.



Po-Yu Chu received the M.S. degree in computer science and information engineering from the National Yunlin University of Science and Technology, Douliu, Yunlin, Taiwan, in 2023.

His research interests include remote sensing image processing and analysis, pattern recognition, computer vision, and machine learning.



Kuan-Liang Liu received the M.S. degree in computer science and information engineering from the National Yunlin University of Science and Technology, Douliu, Yunlin, Taiwan, in 2024.

His research interests include remote sensing image processing and analysis, pattern recognition, computer vision, and machine learning.



Yu-Cheng Wu is currently pursuing the master's degree with Yunlin National University of Science and Technology, Yunlin, Taiwan.

His research interests include remote sensing image processing and analysis, pattern recognition, computer vision, and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:01:23 UTC from IEEE Xplore. Restrictions apply.